

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think> To find the current flowing through the bulb , we can use the formula $P = IV$, where P is the power (in watts), I is the current (in amps), and V is the voltage (in volts). The heat dissipated is given as 40 watts , and the battery is producing 12 volts . We can rearrange the formula to solve for current $I = P / V = 40 / 12 = 3 . 33$ amps </think><answer> 3 . 33 amps , 3 . 6 oh ms </answer>

